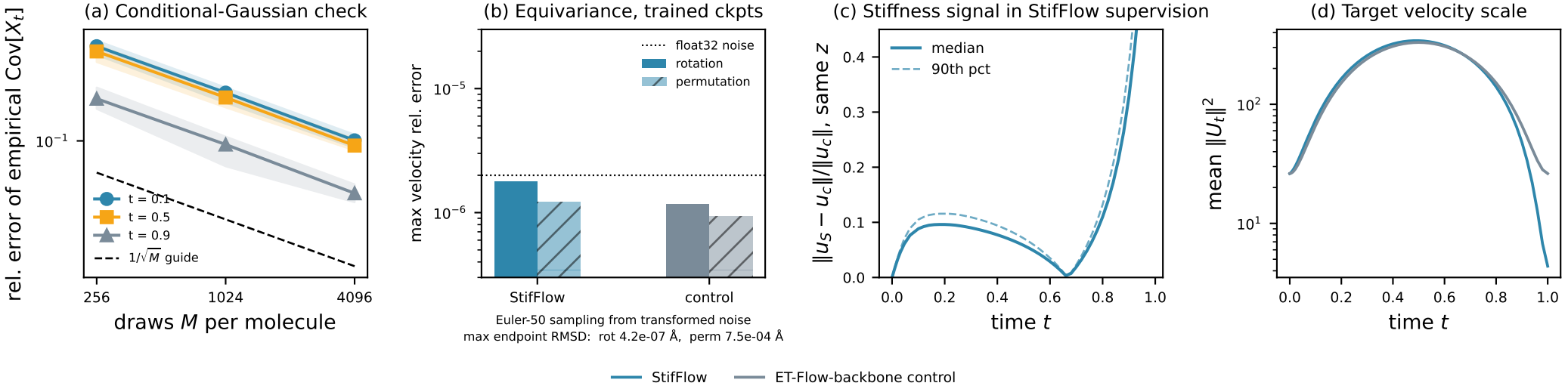
(a) Conditional-Gaussian check
rel. error of empirical Cov[X_t]
10^{-1}
t = 0.1
t = 0.5
t = 0.9
$1/\sqrt{M}$ guide
256
1024
4096
draws M per molecule
(b) Equivariance, trained ckpts
max velocity rel. error
10^{-5}
10^{-6}
float32 noise
rotation
permutation
StifFlow
control
Euler-50 sampling from transformed noise
max endpoint RMSD: rot 4.2e-07 Å, perm 7.5e-04 Å
(c) Stiffness signal in StifFlow supervision
$\|u_S - u_c\|/\|u_c\|$, same z
median
90th pct
0.0
0.1
0.2
0.3
0.4
0.0
0.2
0.4
0.6
0.8
1.0
time t
(d) Target velocity scale
mean $\|U_t\|^2$
10^2
10^1
0.0
0.2
0.4
0.6
0.8
1.0
time t
StifFlow
ET-Flow-backbone control